Chapter 2
Geometric Progression

Real-Life Story (Why Geometric Progression?)

Suppose a bacteria population doubles every hour:

$$2,\ 4,\ 8,\ 16,\ 32,\ldots$$

Each term is obtained by multiplying the previous term by the same number.
Such patterns are called **Geometric Progressions (GP)**.
GP is widely used in population growth, compound interest, computer science, physics, finance, and biology.

1 Geometric Progression

Overview

A **Geometric Progression (GP)** is a sequence in which each term is obtained by multiplying the previous term by a constant value.

This constant value is called the **Common Ratio**.

We will learn:

- Common Ratio
- Types of GP
- nth Term
- Sum of n Terms
- Infinite GP
- Geometric Mean
- Applications of GP

1.1 Definition

A sequence

$$a,\ ar,\ ar^2,\ ar^3,\ldots$$

is called a Geometric Progression if the ratio between consecutive terms remains constant.

$$r = \frac{a_2}{a_1}$$

where:

$$r = \text{Common Ratio}$$

Important Terms

- a is the **first term**
- r is the **common ratio**
- a_n denotes the **nth term**
- S_n denotes the **sum of first n terms**

2 Types of Geometric Progression

1. Increasing GP

If

$$r > 1$$

then the GP increases.

Example:

$$2,\ 4,\ 8,\ 16, \ldots$$

2. Decreasing GP

If

$$0 < r < 1$$

then the GP decreases.

Example:

$$16,\ 8,\ 4,\ 2, \ldots$$

3. Alternating GP

If

$$r < 0$$

then signs alternate.

Example:

$$2,\ -4,\ 8,\ -16,\ldots$$

3 nth Term of GP

Key Formula

$$a_n = ar^{n-1}$$

Solved Example

Find the 6th term of:

$$3,\ 6,\ 12,\ 24,\ldots$$

Given:

$$a = 3, \quad r = 2, \quad n = 6$$

Using:

$$a_n = ar^{n-1}$$

$$a_6 = 3(2)^5$$

$$= 3(32)$$

$$= 96$$

Hence,

$$\boxed{96}$$

4 Sum of First n Terms

Formula:

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

for

$$r \neq 1$$

Solved Example

Find the sum of first 5 terms:

$$2,\ 4,\ 8,\ 16, \ldots$$

Given:

$$a = 2, \quad r = 2, \quad n = 5$$

Using:

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_5 = \frac{2(2^5 - 1)}{2 - 1}$$

$$= \frac{2(32 - 1)}{1}$$

$$= 2(31)$$

$$= 62$$

Hence,

$$\boxed{62}$$

5 Infinite Geometric Progression

If

$$|r| < 1$$

then the sum to infinity exists.
Formula:

$$S_\infty = \frac{a}{1 - r}$$

Solved Example

Find the sum to infinity:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots$$

Given:

$$a = 1, \quad r = \frac{1}{2}$$

Using:

$$
\begin{aligned}
S_\infty &= \frac{a}{1-r} \\
&= \frac{1}{1-\frac{1}{2}} \\
&= \frac{1}{\frac{1}{2}} \\
&= 2
\end{aligned}
$$

Hence,

$$\boxed{2}$$

6 Geometric Mean

If $a,\ G,\ b$ are in GP, then:

$$G = \sqrt{ab}$$

Solved Example

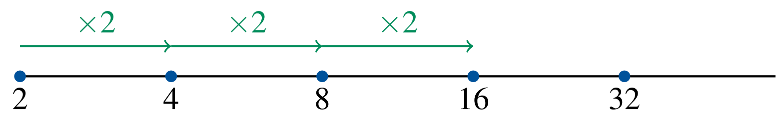
Find the geometric mean between 4 and 16.

$$
\begin{aligned}
G &= \sqrt{4 \times 16} \\
&= \sqrt{64} \\
&= 8
\end{aligned}
$$

Hence,

$$\boxed{8}$$

7 Visual Understanding



8 Applications of GP

- Compound Interest
- Population Growth
- Radioactive Decay
- Computer Algorithms
- Bacteria Growth
- Financial Investments

9 Solved Questions

Solved Question 1

Find the common ratio of:

$$3,\ 9,\ 27,\ 81,\ldots$$

Solution

$$r = \frac{9}{3} = 3$$

Hence,

$$\boxed{3}$$

Solved Question 2

Find the 8th term of:

$$2,\ 6,\ 18,\ldots$$

Solution

$$a = 2, \quad r = 3, \quad n = 8$$

$$a_8 = 2(3)^7$$

$$= 2(2187)$$

$$= 4374$$

Hence,

$$\boxed{4374}$$

Solved Question 3

Find the sum of first 6 terms:

$$1,\ 2,\ 4,\ 8,\ldots$$

Solution

$$a=1,\quad r=2,\quad n=6$$

$$S_6=\frac{1(2^6-1)}{2-1}$$

$$=64-1$$

$$=63$$

Hence,

$$\boxed{63}$$

Solved Question 4

Find the geometric mean between 9 and 25.

Solution

$$G=\sqrt{9\times 25}$$

$$=\sqrt{225}$$

$$=15$$

Hence,

$$\boxed{15}$$

Solved Question 5

Find the sum to infinity:

$$3+\frac{3}{2}+\frac{3}{4}+\ldots$$

Solution

$$a=3,\quad r=\frac{1}{2}$$

$$S_\infty=\frac{3}{1-\frac{1}{2}}$$

$$= \frac{3}{\frac{1}{2}}$$

$$= 6$$

Hence,

$$\boxed{6}$$

10 Practice Questions

Basic Level

1. Find the common ratio:
$$5,\ 15,\ 45,\ldots$$
2. Find the 7th term:
$$2,\ 4,\ 8,\ldots$$
3. Find the sum of first 5 terms:
$$1,\ 3,\ 9,\ldots$$
4. Find the geometric mean between 4 and 64.
5. Determine whether:
$$2,\ 6,\ 18,\ldots$$
is a GP.

Advanced Level

6. Find the sum to infinity:
$$1+\frac{1}{3}+\frac{1}{9}+\ldots$$
7. Which term of the GP:
$$3,\ 6,\ 12,\ldots$$
is equal to 192?
8. Insert 2 geometric means between 3 and 81.
9. If the 4th term is 54 and common ratio is 3, find first term.
10. Find the sum:
$$2+4+8+\cdots+1024$$

11 Quick Formula Sheet

Concept	Formula	Meaning
nth Term	$a_n = ar^{n-1}$	General term
Sum of n Terms	$S_n = \frac{a(r^n - 1)}{r - 1}$	Finite GP sum
Sum to Infinity	$S_\infty = \frac{a}{1 - r}$	Infinite GP sum
Geometric Mean	$G = \sqrt{ab}$	Middle term

Mathematics Becomes Easy with Practice